Configuring and deploying AI agents

About prompting Atlas AI agents

Copy MCP Server

Learn prompting best practices to get accurate and relevant responses from Atlas AI agents for your industrial data challenges.

Atlas AI agents use language models to solve industrial problems by querying your knowledge graph. The quality of your prompts directly affects the accuracy and usefulness of agent responses. Effective prompts help agents understand what you need and deliver relevant information for tasks like maintenance planning, equipment monitoring, and operational decision-making.

Effective prompts can also help produce accurate results faster, reduce the need for follow-up questions, and help you get more value from your industrial data.

How agent configuration affects prompting

When you build agents, your configuration settings affect how they interpret and respond to your prompts.

Language models

The language model you select affects how agents interpret your prompts and generate responses. Different models have different strengths for different types of tasks.

Select a language model that matches your use case and response time needs. More advanced models can provide more detailed responses, but can respond more slowly than less advanced models. See **language models** for the list of available models.

- **Instructions** define how to achieve the goal and specify the workflow, scope, and approach.

Effective instructions are specific and focused. General instructions apply to all agent actions, while tool-specific instructions guide behavior for specific tasks.

Learn more about configuring agent instructions in **Build and publish agents**.

Tools

Agents use tools to access Cognite Data Fusion (CDF) data, run calculations, or integrate with external systems. The tools you enable control what tasks agents can perform. When building agents, select only the tools that you need for your use case to help improve agent response quality.

See **available tools** for descriptions of each tool you can enable.

Writing effective prompts

Apply the following principles to write prompts that can produce accurate, relevant responses.

Be specific and clear

Specific prompts can produce better results than vague ones. Use action verbs to clarify what you want the agent to do.

Vague Specific Highly specific

This vague prompt doesn't give the agent enough information to focus the response. The agent can't determine whether you need operational status, maintenance history, specifications, or location information, which can result in irrelevant or incomplete responses.

Provide context

Context helps agents focus on relevant data by narrowing the scope of your request.

Without context | With basic context | With detailed context

```
Show me the temperature trends.
```

Without knowing which equipment, sensor, or time period to analyze, this prompt leaves the agent searching through irrelevant data. You might receive temperature trends for the wrong equipment or an unhelpful time range.

Define the desired output

Specify the format, structure, and level of detail you need in the agent response.

Undefined | Defined format | Defined precision | Structured output

```
Give me maintenance data for Valve V-789.
```

Understanding agent responses

When agents produce unexpected results, Atlas AI provides insight into their decision-making process to help you refine your prompts.

Analyzing the reasoning field

Atlas AI displays agent decision-making in a reasoning field that shows how agents understand prompts, select tools, and process information.

Use the reasoning field to understand agent behavior and refine your approach.

- Check if agents follow your specified workflow or deviate from expected steps
- Identify how agents interpret ambiguous terms or instructions
- Review why agents select specific tools at decision points
- Observe what information guides agent responses and decision-making

When the reasoning reveals gaps between your intent and the agent's interpretation, adjust your prompt to provide clearer direction or a more specific context.

Reviewing tool calls

Tool call inputs and outputs reveal how agents translate your prompts into actions.

Review tool calls to understand how agents execute your requests.

- Verify that all required parameters are included and formatted correctly
- Confirm the agent selected appropriate tools for your request
- Observe how agents interpret and use the tool results in their responses
- Check how agents handle tool failures or unexpected outputs

Building on conversation history

Your conversations with agents are saved as records that you can use to build knowledge for more confident decisions in your industrial operations. You can use these records of past conversations to verify insights, reuse prompting approaches, and support compliance.

Verifying insights

Review the reasoning in saved conversations to understand how your agents generate responses. You can use this insight to evaluate your agent's data retrieval methods and refine your prompts.

Reusing prompts

You can review conversation history to identify which prompting approaches produce the results you need for your tasks. Use these approaches as templates by applying the same patterns to similar tasks.

Supporting compliance

Industrial operations require evidence of how decisions are made. Conversation history stores your agent's reasoning and the data it references so you can trace decision-making for compliance reports or safety audits.

Related information

- Build and publish agents
- Evaluate agents
- Language models

Was this page helpful?

Yes No

Build and publish agents
Previous

Language model library
Next